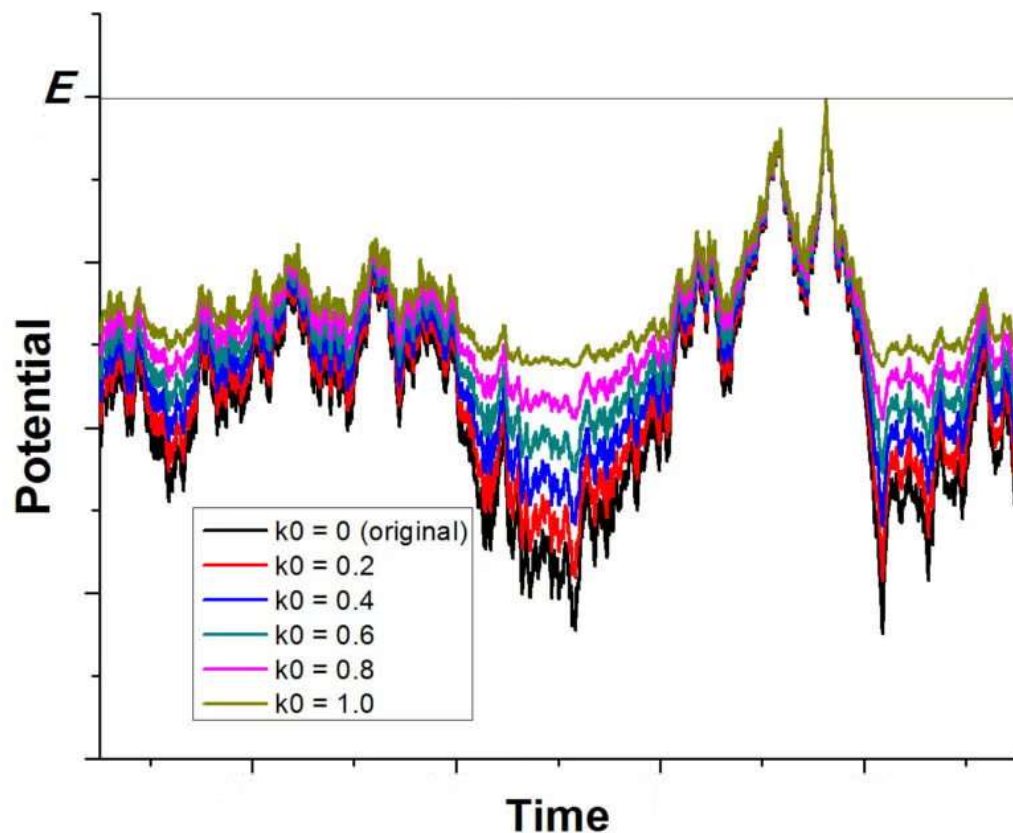


Gaussian accelerated Molecular Dynamics

Naf Guo
2026



与metadynamics的一大区别是不用自己定义CV

$$V^*(r) = V(r) + \Delta V(r),$$

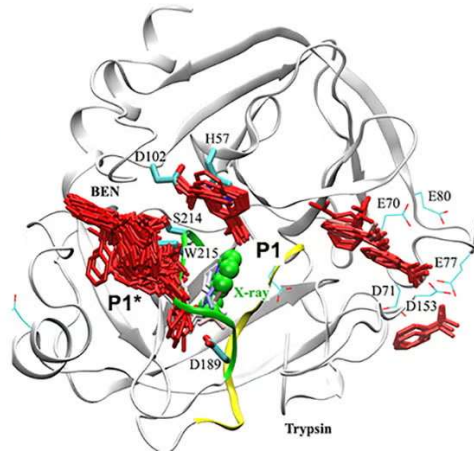
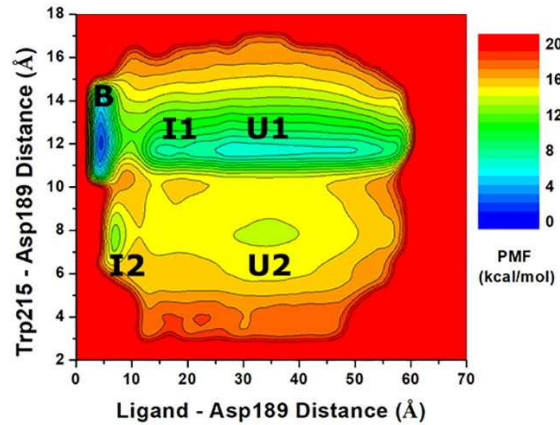
$$\Delta V(r) = \begin{cases} \frac{1}{2}k(E - V(r))^2, & V(r) < E \\ 0, & V(r) \geq E \end{cases}$$

GaMD 通过给系统加一个增强势（boost potential），主动抬高低势能区域，从而平滑势能面、降低势垒高度，让系统更容易跨越势垒、探索更多构象，同时保证后续能通过重权重恢复真实的自由能信息。

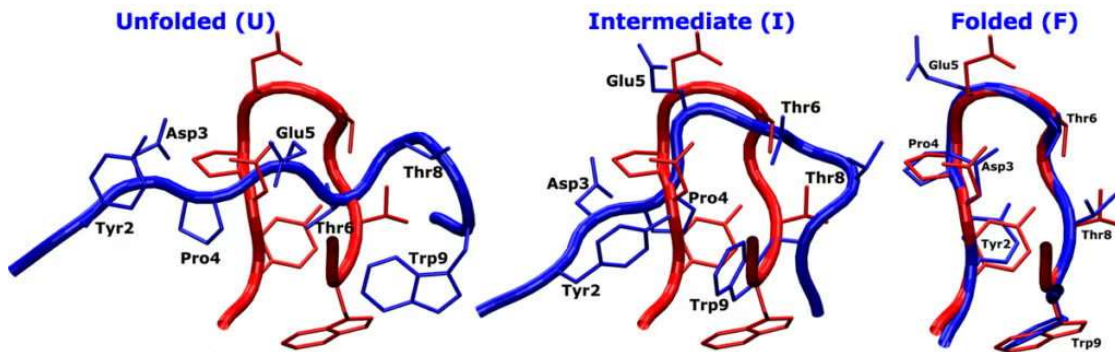
GaMD - examples

<https://www.med.unc.edu/pharm/miaolab/resources/gamd/examples/>

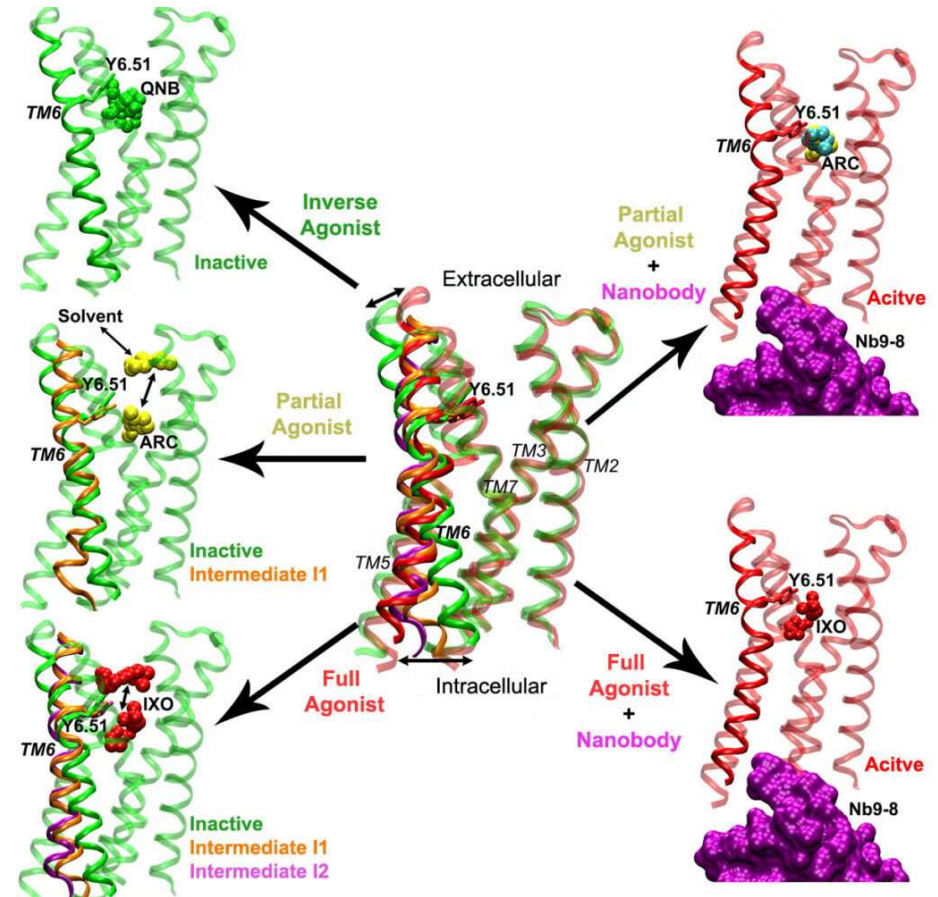
Protein-Ligand binding



Protein folding



Conformation distribution upon ligand binding



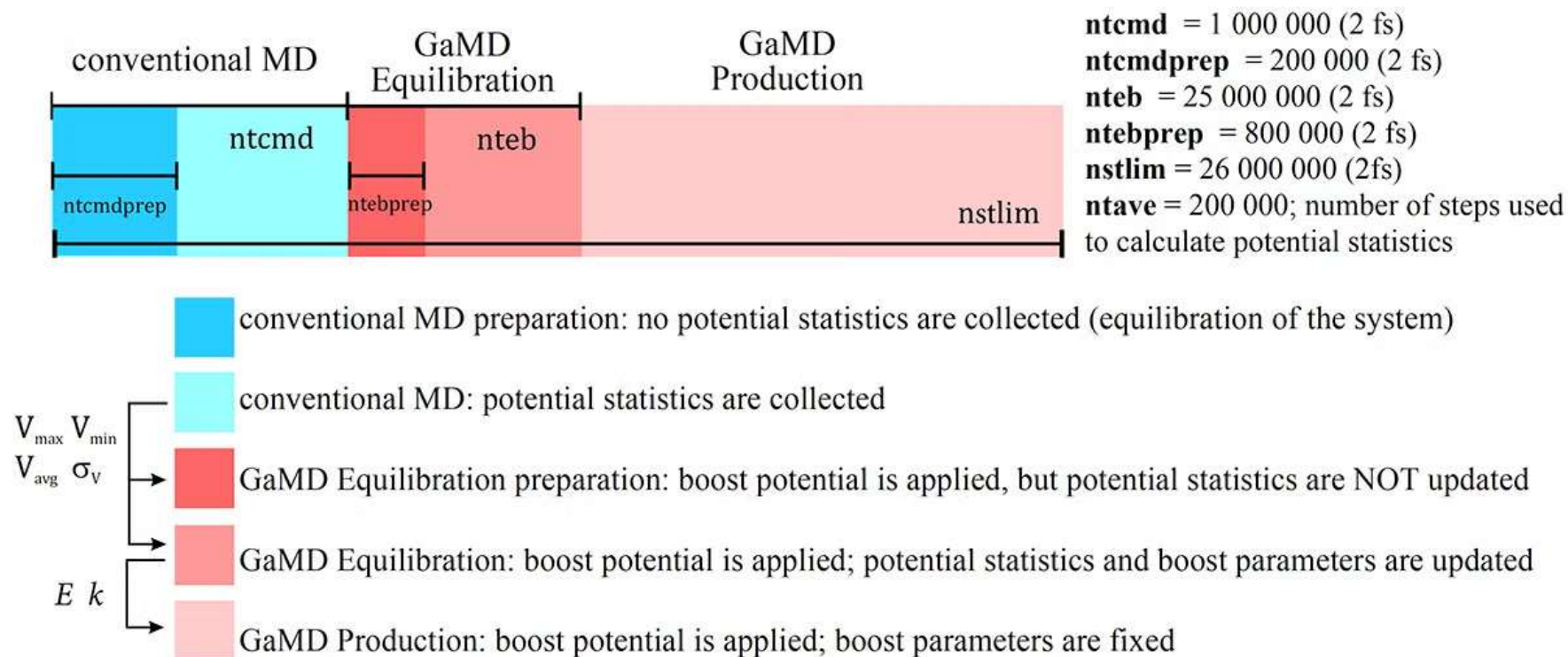
GaMD - parameters

<https://www.med.unc.edu/pharm/miaolab/resources/gamd/manual/>

常规 MD 阶段：先弛豫稳定体系，再收集势能统计数据，为增强势参数计算提供基准。

GaMD 平衡阶段：先让体系适应初始增强势，再动态校准增强参数，得到稳定设置。

GaMD 生产阶段：使用固定的增强参数运行长时间模拟，收集轨迹数据，用于后续重权重分析以恢复真实自由能面。



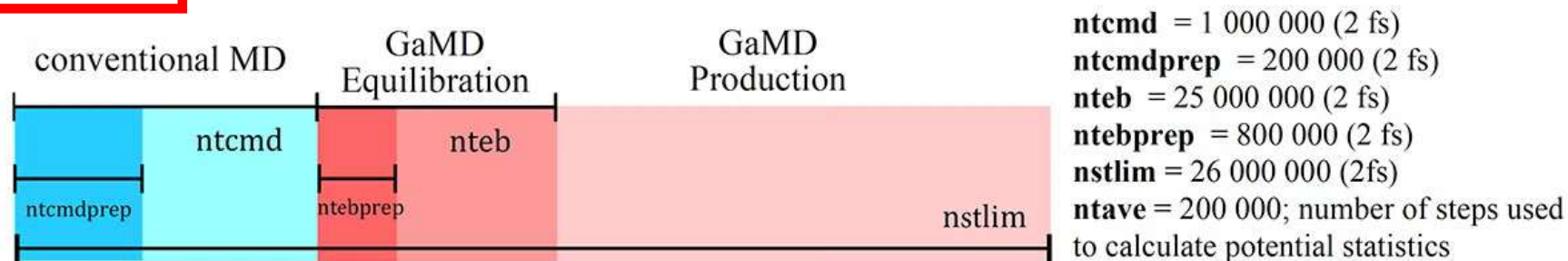
GaMD - parameters

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需要在普通模拟的Mdin文件的基础上修改部分参数以及加入额外的GaMD参数

Job #1: cMD and GaMD equilibration

`nstlim = 26000000,`

`irest = 0,`

`ntx = 1,`

`igamd = 3, iE = 1, irest_gamd = 0,`

`ntcmd = 1000000, nteb = 25000000, ntave = 200000,`

`ntcmdprep = 200000, ntebprep = 800000,`

`sigma0P = 6.0, sigma0D = 6.0,`

`ntcmdprep=200000`

`ntcmd = 1000000`

常规MD的平衡部分，包含在ntcmd中，这部分时间不会用来收集数据

常规MD 总时间2ns，与ntcmdprep的差值时间会用于收集数据，用于计算增强势参数

<https://www.med.unc.edu/pharm/miaolab/resources/gamd/tutorial/>

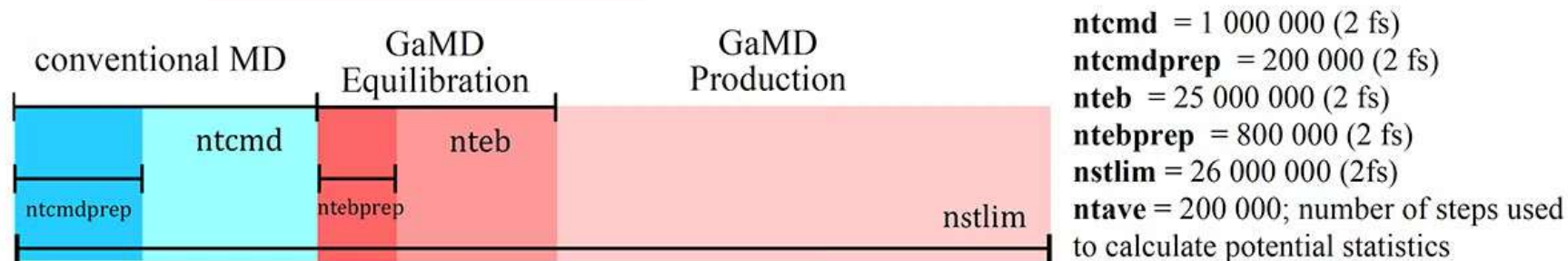
GaMD - parameters

<https://www.med.unc.edu/pharm/miaolab/resources/gamd/manual/>

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Job #1: cMD and GaMD equilibration

nstlim = 26000000,

irest = 0,

ntx = 1,

igamd = 3, **iE** = 1, **irest_gamd** = 0,

ntcmd = 1000000, **nteb** = 25000000, **ntave** = 200000,

ntcmdprep = 200000, **ntebprep** = 800000,

sigma0P = 6.0, **sigma0D** = 6.0,

ntebprep=800000

nteb = 1000000

GaMD平衡阶段的弛豫部分，包含在nteb中，这部分时间不会更新增强势参数

GaMD平衡阶段，与ntebprep的差值时间会用于调整增强势参数

<https://www.med.unc.edu/pharm/miaolab/resources/gamd/tutorial/>

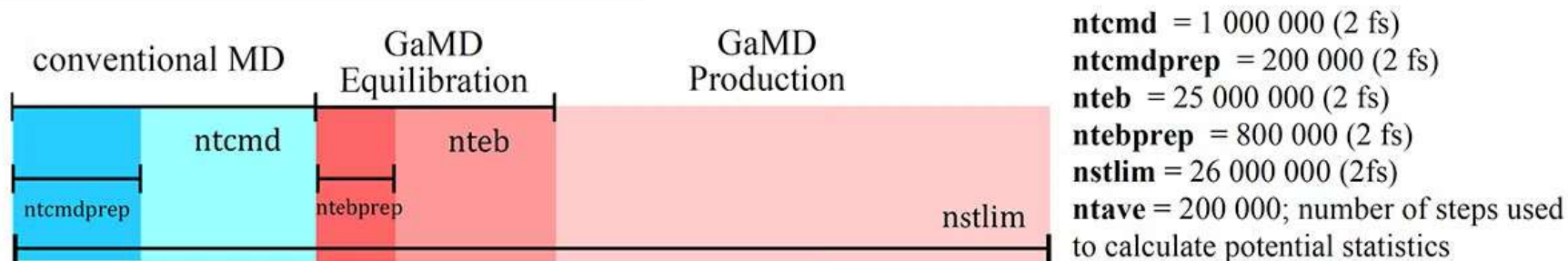
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Job #1: cMD and GaMD equilibration

nstlim = 26000000,

irest = 0,

ntx = 1,

igamd = 3, iE = 1, irest_gamd = 0,

ntcmd = 1000000, nteb = 25000000, ntave = 200000,

ntcmdprep = 200000, ntebprep = 800000,

sigma0P = 6.0, sigma0D = 6.0,

ntcmd = 1000000

常规MD

nteb = 1000000

GaMD平衡阶段

可以注意带这个设置里 $nstlim = ntcmd + nteb$ ，所以这个设置对应的 GaMD production 时间为 0，也就是为什么这个控制文件只是用来完成两个平衡阶段

<https://www.med.unc.edu/pharm/miaolab/resources/gamd/tutorial/>

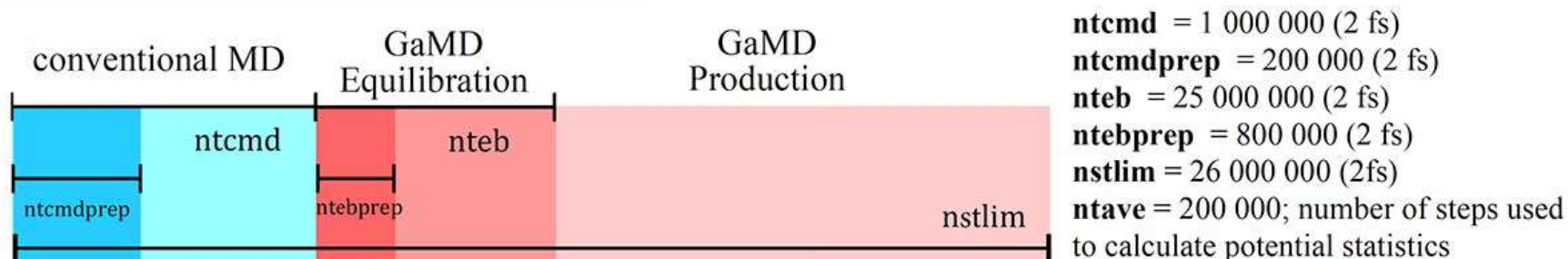
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Job #1: cMD and GaMD equilibration

关键参数 **ntave**

nstlim = 26000000,

irest = 0,

ntx = 1,

igamd = 3, **iE** = 1, **irest_gamd** = 0,

ntcmd = 1000000, **nteb** = 25000000, **ntave** = 200000,

ntcmdprep = 200000, **ntebprep** = 800000,

sigma0P = 6.0, **sigma0D** = 6.0,

ntave代表每隔多少步会收集势能统计数据（在常规MD阶段），以及进一步校正增强参数（GaMD平衡阶段）
推荐值：约为体系**总原子数的 4 倍**（如 10,000 原子 → 40,000 步）。

约束：**ntcmd**、**nteb** 必须是 **ntave** 的整数倍，保证统计更新同步。

igamd=3, **iE**=1告诉amber现在跑的是GaMD

irest_gamd=0 代表新模拟，需要产生GaMD增强势参数存在"gamd-restart.dat"

<https://www.med.unc.edu/pharm/miaolab/resources/gamd/tutorial/>

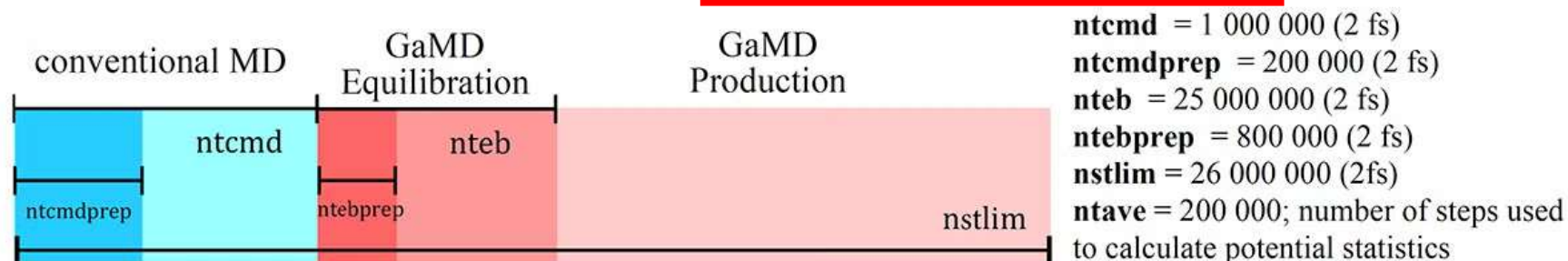
GaMD - parameters

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GaMD 生产阶段：使用固定的增强参数运行长时间模拟，收集轨迹数据，用于后续重权重分析以恢复真实自由能面。



Job #2: start GaMD production simulation

`nstlim = 50000000,`

`irest = 0,`

`ntx = 1,`

`igamd = 3, iE = 1, irest_gamd = 1,`

`ntcmd = 0, nteb = 0, ntave = 200000,`

`ntcmdprep = 0, ntebprep = 0,`

`sigma0P = 6.0, sigma0D = 6.0,`

`irest=0`是amber本身的参数，代表不会使用之前模拟产生的原子速度
`ntx=1`从inpcrd文件中只读入坐标，不读入速度，与`irest=0`匹配

`igamd=3`，`iE=1`告诉amber现在跑的是GaMD

`irest_gamd=1`代表不是新模拟，需要从已有的“gamd-restart.dat”读取GaMD参数

注意到`ntcmd`和`nteb`均为0，说明这个文件完全不跑平衡，只是GaMD生产阶段

<https://www.med.unc.edu/pharm/miaolab/resources/gamd/tutorial/>

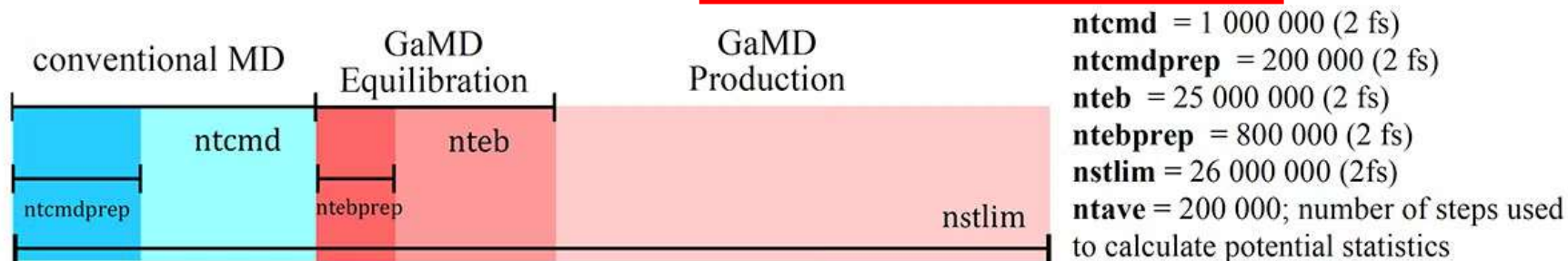
GaMD - parameters

<https://www.med.unc.edu/pharm/miaolab/resources/gamd/manual/>

常规 MD 阶段：先弛豫稳定体系，再收集势能统计数据，为增强势参数计算提供基准。

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Job #3: repeat running jobs using this input file until end of your GaMD production simulation

```
nstlim = 50000000,
```

```
irest = 1,
```

```
ntx = 5,
```

```
igamd = 3, iE = 1, irest_gamd = 1,
```

```
ntcmd = 0, nteb = 0, ntave = 200000,
```

```
ntcmdprep = 0, ntebprep = 0,
```

```
sigma0P = 6.0, sigma0D = 6.0,
```

Note: You can also perform multiple independent GaMD simulations with randomized atomic velocities by repeating jobs #2 and #3.

irest=1代表使用之前模拟产生的原子速度
ntx=5 读取坐标和速度，与**irest=1**匹配

这是你续跑GaMD生产相的时候用的控制文件

注意以上列出的控制文件都是要在一个标准的amber控制文件的基础上添加的

<https://www.med.unc.edu/pharm/miaolab/resources/gamd/tutorial/>

GaMD - parameters

<https://www.med.unc.edu/pharm/miaolab/resources/gamd/manual/>

<code>igamd</code>	Flag to apply boost potential = 0 (default) no boost is applied = 1 boost on the total potential energy only = 2 boost on the dihedral energy only = 3 dual boost on both dihedral and total potential energy = 4 boost on the non-bonded potential energy only = 5 dual boost on both dihedral and non-bonded potential energy = 10 boost on non-bonded potential energy of selected region (defined by <code>timask1</code> and <code>scmask1</code>) as for a ligand (LiGaMD) = 11 dual boost on both non-bonded potential energy of the bound ligand and remaining potential energy of the rest of the system (LiGaMD_Dual)	<code>ntcmdprep</code> The number of preparation conventional molecular dynamics steps. This is used for system equilibration and the potential energies are not collected for calculating their statistics. The default is 200,000 for a simulation with 2 fs timestep. <code>ntcmd</code> The number of initial conventional molecular dynamics simulation steps. Potential energies are collected between <code>ntcmdprep</code> and <code>ntcmd</code> to calculate their maximum, minimum, average and standard deviation (V_{max} , V_{min} , V_{avg} , σ_V). The default is 1,000,000 for a simulation with 2 fs timestep. <code>ntebprep</code> The number of preparation biasing molecular dynamics simulation steps. This is used for system equilibration after adding the boost potential and the potential statistics (V_{max} , V_{min} , V_{avg} , σ_V) are not updated during these steps. The default is 200,000 for a simulation with 2 fs timestep. <code>nteb</code> The number of biasing molecular dynamics simulation steps. Potential statistics (V_{max} , V_{min} , V_{avg} , σ_V) are updated between the <code>ntebprep</code> and <code>nteb</code> steps and used to calculate the GaMD acceleration parameters, particularly E and k_0 . The default is 1,000,000 for a simulation with 2 fs timestep. A greater value may be needed to ensure that the potential statistics and GaMD acceleration parameters level off before running production simulation between the <code>nteb</code> and <code>nstlim</code> (total simulation length) steps. Moreover, <code>nstlim</code> can be set to <code>ntcmd+nteb</code> , by which the potential statistics and GaMD acceleration parameters are updated adaptively throughout the simulation. This in some cases provides more appropriate acceleration.
<code>iE</code>	Flag to set the threshold energy E = 1 (default) set the threshold energy to the lower bound $E = V_{max}$ = 2 set the threshold energy to the upper bound $E = V_{min} + (V_{max} - V_{min})/k_0$	<code>ntave</code> The number of simulation steps used to calculate the average and standard deviation of potential energies. This variable has already been used in Amber. The default is set to 50,000 for GaMD simulations. It is recommended to be updated as about 4 times of the total number of atoms in the system. Note that <code>ntcmd</code> and <code>nteb</code> need to be multiples of <code>ntave</code> .
<code>iEP</code>	Flag to overwrite <code>iE</code> and set the threshold energy E for applying the first boost potential in dual-boost schemes = 1 (default) set the threshold energy to the lower bound $E = V_{max}$ = 2 set the threshold energy to the upper bound $E = V_{min} + (V_{max} - V_{min})/k_0$	<code>ntave</code> 是 GaMD 中用于计算势能平均值和标准差的统计窗口步数。
<code>iED</code>	Flag to overwrite <code>iE</code> and set the threshold energy E for applying the second boost potential in dual-boost schemes = 1 (default) set the threshold energy to the lower bound $E = V_{max}$	推荐值：约为体系总原子数的 4 倍（如 10,000 原子 → 40,000 步）。 约束： <code>ntcmd</code> 、 <code>nteb</code> 必须是 <code>ntave</code> 的整数倍，保证统计更新同步。

GaMD - Tutorial

<https://www.med.unc.edu/pharm/miaolab/resources/gamd/tutorial/>

Note: While the NVT ensemble has been used in the tutorial simulations with “ntb=1”, the NPT ensemble is generally preferred for running GaMD simulations (especially for the GaMD equilibration) with parameters like “ntb = 2, ntp = 1, barostat = 1”.

Miao Lab

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Resources

DBMD

GaMD

Theory

Analysis

Examples

GaMD-Amber: Installation

GaMD-Amber: Manual

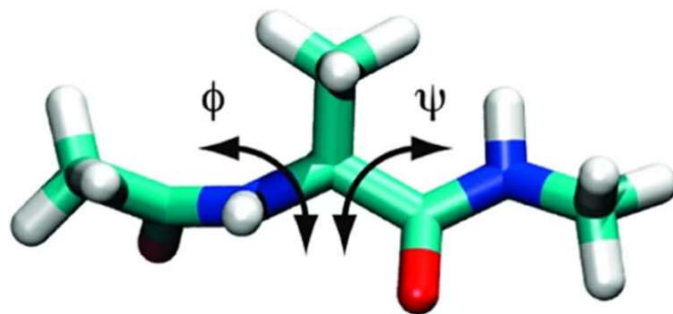
GaMD-Amber: Tutorial

GaMD-NAMD: Installation

GaMD-NAMD: Manual

GaMD-Amber: Tutorial

Alanine Dipeptide



[Download a PDF tutorial of running GaMD simulations on alanine dipeptide.](#)

[Download Source files of the GaMD-Amber tutorial](#) (31.8MB zip file)

GaMD for membrane protein

https://pmc.ncbi.nlm.nih.gov/articles/instance/10685416/bin/au3c00503_si_001.pdf

Example input file for GaMD production simulations of GPCRs without G-protein bound

```
GaMD production
&cntrl
  imin=0,      ! No minimization
  irest=0,     ! This IS a new MD simulation
  ntx=1,       ! read coordinates only

  ! Temperature control
  ntt=3,       ! Langevin dynamics
  gamma_ln=1.0, ! Friction coefficient (ps^-1)
  tempi=310.0,  ! Initial temperature
  temp0=310.0, ! Target temperature
  ig=-1,       ! random seed

  ! Potential energy control
  cut=9.0,     ! nonbonded cutoff, in Angstroms

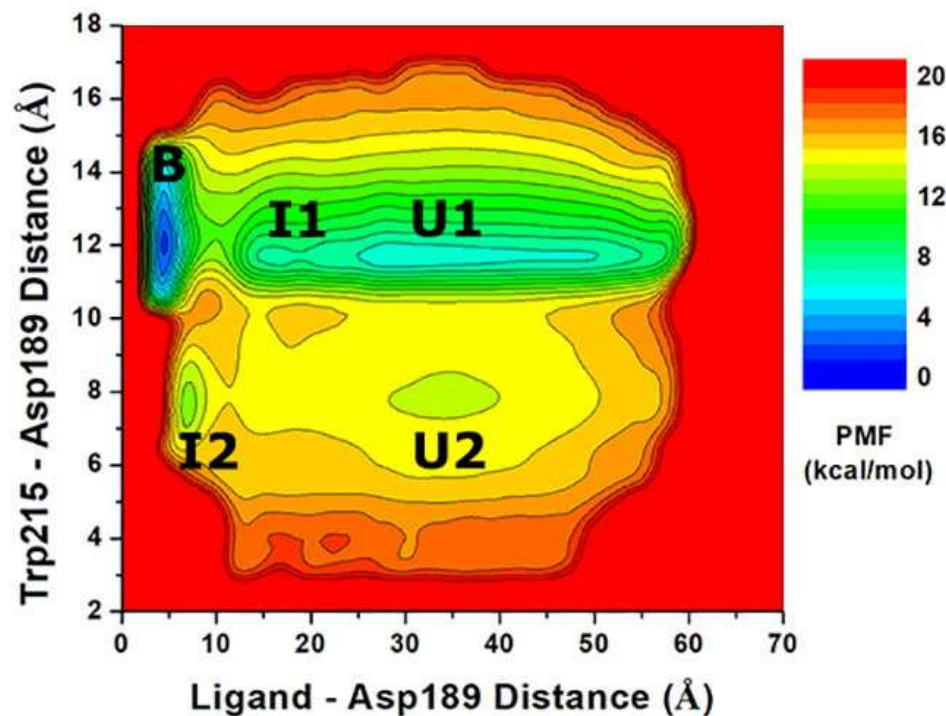
  ! MD settings
  nstlim=250000000, ! simulation length
  dt=0.002,      ! time step (ps)

  ! SHAKE
  ntc=2,        ! Constrain bonds containing hydrogen
  ntf=1,        ! Do not calculate forces of bonds containing hydrogen
```


Reweighting

<https://www.med.unc.edu/pharm/miaolab/resources/pyreweighting/>

<https://github.com/MiaoLab20/pyreweighting>



GaMD是一种增强采样方法，它是改变了自由能面的形状的，为了获得正确的自由能面，需要做reweighting，Miao实验室提供了对应的脚本。

Workshop Tutorials

The GaMD workshop will provide “hands-on” tutorial sessions different GaMD algorithms, including the GaMD, Ligand GaMD 3 (LiGaMD3) and Protein-Protein Interaction GaMD (PPI-GaMD). You can download the tutorial with the required simulation and analysis files in compressed tar.bz2 files at:

GaMD tutorial: [GaMD-Tutorial-2024.pdf](#) and [GaMD.tar.bz2](#) (24.5MB)

LiGaMD3 tutorial: [LiGaMD3-Tutorial-2024.pdf](#) and [LiGaMD3.tar.bz2](#) (33.7MB)

PPI-GaMD tutorial: [PPI-GaMD-2024.pdf](#) and [PPI-GaMD.tar.bz2](#) (16MB)

PyReweighting: [PyReweighting24.tar.bz2](#) (12KB)

FAQ

Frequently Asked Questions about attending the GaMD workshop.